

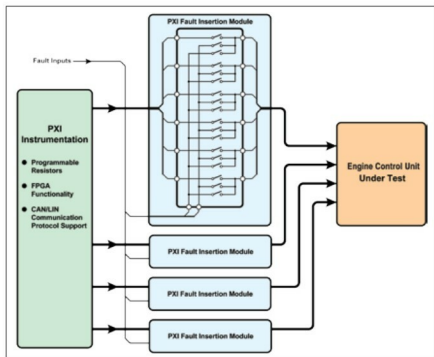


Fig. 2: Block diagram of HILS test application

Here we discuss several applications where PXI was an important part of the required test strategy and in some cases made the efforts easier. Some of these applications are unique enough and one might think these would not require advanced testing techniques, much less any simulation. But in each case, PXI was a major portion of the test strategy.

1 Automotive ECU fault insertion

Customer. A leading French systems integrator who develops functional and design validation testers for the automotive and aerospace industries.

Application. The end-user required test systems to validate the design reliability of their engine control units—also known as power-train control modules or engine control modules.

The system integrator was tasked to find a more flexible and cost-effective alternative to proprietary HILS solutions. The existing system was an internally-developed design and used a very manual method of performing fault simulation. To insert

a fault, the operator had to manually move cables on a patch panel to short pins together, force a stuck at Vbat or ground, or inject false sensor data. This method was slow, subject to human error and expensive. Given the issue of liability when electronic systems fail and the continuous pressure to lower costs, a strategy change was clearly necessary.

Fault simulation during the design and validation of ECUs is a method of establishing solid predictions for reliability and firmware verification, ensuring the safety of driver and passengers. An ECU relies on information from a set of sensors and controls to decide what to do with the device that it is managing. These sensors often work in extremely hostile environments (such as within a car in this application) and predictable failures can occur in the sensors or their interconnections. The ECU has to respond appropriately to these component failures, as well as to genuine system faults.

The idea of testing for system failures is not new. It is an important aspect of ECU validation and involves

the introduction of electrical faults into a system. The test process typically duplicates various conditions that could occur because of corrosion, short/open circuits and other electrical failures inherited through infant mortality, age, damage or even faulty installation.

Solution. PXI provides an open platform for HILS requirements. Combining this with the large range of hardware available from vendors enables flexible and cost-effective alternatives to proprietary systems.

The modularity and openness of PXI enabled the integrator to design a highly scalable solution with plenty of potential for evolution.

There is a wide range of fault-insertion switching modules in PXI, specifically designed to assist the simulation of fault conditions in automotive and avionics applications that involve the reliability testing of safety-critical controllers. The block diagram (Fig. 2) shows how fault-insertion modules work in this test strategy. These switching modules are designed to route various fault conditions between a test fixture and the UUT, which include open circuit, short circuit between UUT connections and short circuits to other signals such as power, ignition and ground. Faulty serial buses can be injected into the circuit as well.

There are many fault-simulation systems that are available commercially. However, these tend to be proprietary to one vendor and might not work well with other vendors' products, and can be very expensive as well.

Another key point in the selection of the PXI standard was the fact that every vendor offered software-driver support for real-time operating systems (RTOS)—a key requirement in HILS. The reason for RTOS is the deterministic nature of the test. Responses by the UUT, especially in the areas of active safety, need to be presented at exact times in order to protect the vehicle's passengers.

The solution was accommodated in four 3U PXI chassis. Modular products